5. Evidence Preservation and Analysis

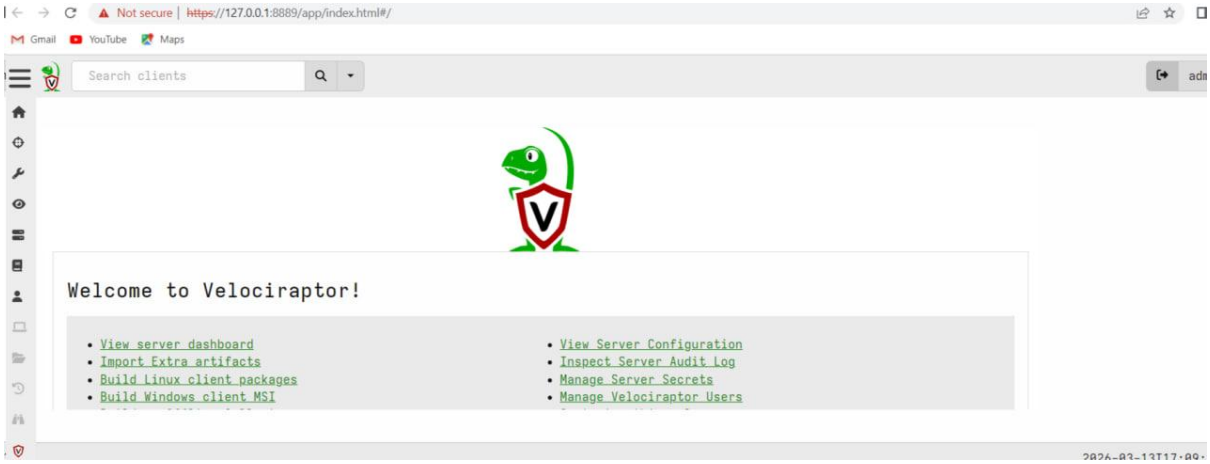
The objective of this task is to collect volatile system evidence from a Windows endpoint and preserve it securely for forensic analysis. The evidence collected includes network connection data and a system memory dump. The integrity of the evidence is verified using cryptographic hashing, and proper chain-of-custody documentation is maintained.

Environment

- Component — Description
- DFIR Tool — Velociraptor
- Forensic Tool — FTK Imager
- Endpoint — Windows Virtual Machine
- Investigator — SOC Analyst

1. Volatile Data Collection

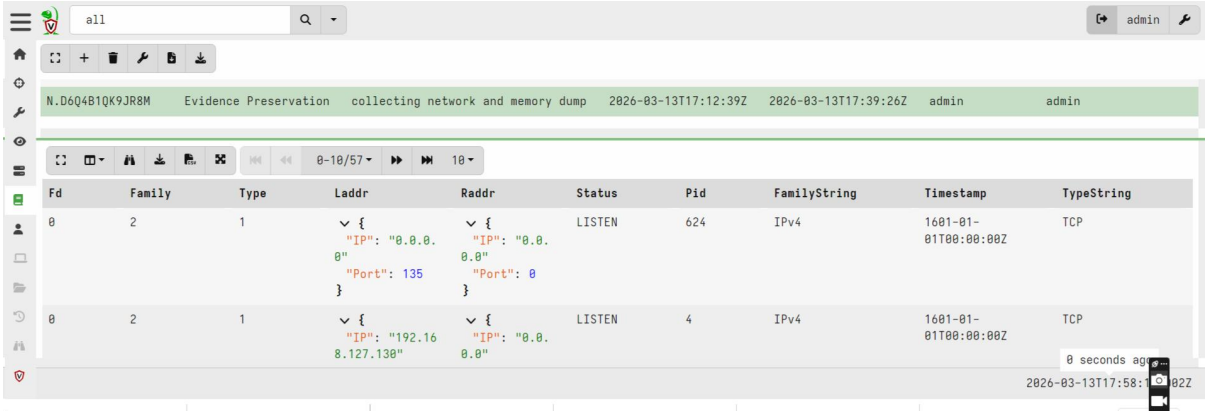
Volatile data refers to system information that exists only while the system is running. This includes active network connections, running processes, and system memory. Collecting volatile data helps investigators understand the system's state during a potential security incident.



Procedure

The Velociraptor client was used to collect network connection information from the Windows endpoint.
The following query was executed using Velociraptor Query

SELECT * FROM netstat



This query retrieves information about active network connections, including:

- Local IP addresses
- Remote IP addresses
- Open ports
- Connection states

The collected results were exported and saved in CSV format for further analysis and documentation.

2. Memory Evidence Collection

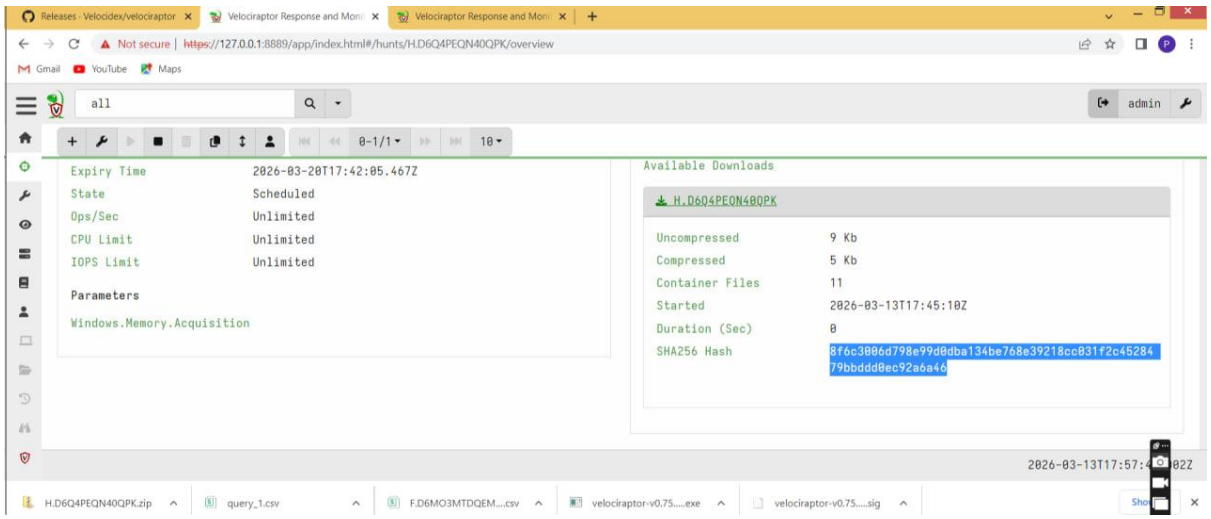
Memory acquisition captures the contents of system RAM, which may contain important forensic artifacts such as:

- Running processes
- Encryption keys
- Network connections
- Malware artifacts
- Command execution history

This data is valuable during incident response investigations.

Using Velociraptor, the following artifact query was executed to acquire a memory dump:

SELECT * FROM Artifact.Windows.Memory.Acquisition



This artifact collected the full system memory from the Windows virtual machine and stored it as a forensic memory image.
The memory dump file was preserved for further forensic analysis.

3. Evidence Integrity Verification

To ensure that the collected evidence was not altered, a cryptographic hash value was generated using the SHA-256 hashing algorithm.
The following command was used to generate the hash value:

```
sha256sum memory_dump.raw
```

Hashing ensures evidence integrity and helps investigators verify that the evidence remains unchanged throughout the investigation process.

4. Chain of Custody Documentation

A chain-of-custody record was maintained to document the collection and handling of digital evidence.

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-Y Memory Dump	SOC Analyst	3/13/2026	SHA256_HASH_VALUE

This documentation ensures that the evidence remains legally valid and traceable during the investigation process.

4. Result

The volatile network connection data and system memory dump were successfully collected using Velociraptor. The collected evidence was securely preserved and its integrity verified using SHA-256 hashing. All evidence collection activities were documented through chain-of-custody records to ensure proper forensic handling.

Evidence preservation is a critical step in digital forensics and incident response. By collecting volatile data and memory artifacts using Velociraptor and verifying their integrity through hashing, investigators can maintain reliable evidence for further analysis and incident investigation.